

EXP → V5 Cheat Sheet

Pull this out when you join the team

Everything you learned on EXP, in the words the V5 team uses. A few names change; the thinking does not. **A VEX match = an autonomous period + a driver period — and you already do both (your Sumo auto-hunt and your tank driving).**

Concept	A&R · VEX EXP (VEXcode)	Team · VEX V5 (SpartanDrive)
Robot's computer	EXP Brain — built-in inertial sensor	V5 Brain — built-in IMU
The motors	Smart Motor — 200 RPM, fixed	Smart Motor — swappable cartridges
Bundle a side's motors	motor_group leftSide = motor_group(...)	the left/right groups SpartanDrive uses
Whole-robot drive object	a (smart) drivetrain	the drive object
Tank (two-stick) driving	left & right motor_group from the sticks	drive.tank(leftY, rightY)
Arcade (one-stick) driving	Drivetrain.arcade(power, turn)	drive.arcade(power, turn)
Drive a set distance	Drivetrain.driveFor(forward, 24, inches)	drive.drive_distance(24)
Turn by an angle	Drivetrain.turnFor(right, 90, degrees)	drive.turn_to_angle(90)
Turn to face a heading	Drivetrain.turnToHeading(90, degrees)	drive.turn_to_angle(90)
Which way am I facing?	Brain inertial → heading	drive.heading()
Where am I on the field?	— not tracked on EXP —	odometry → drive.get_pose()
Ready the sensor	inertial calibrates at Brain startup	drive.calibrate()
Run-by-itself routine	your Sumo auto-hunt	autonomous()
Driver loop (1 drive call/pass)	while (true) { Drivetrain.arcade(...) }	opcontrol() { ... drive.tank(...) }
Where you write & run code	VEXcode (VS Code) → Download & Run	PROS — different editor & layout
The C++ language & logic	same thinking	same thinking

The one line: same C++ thinking, same robot concepts — a handful of methods get renamed and the editor changes when you reach the team. You're not starting over; you're picking up where this course leaves off.